WORKED SOLUTION

STEP III 2007, Question 14

Probability and Statistics • Geometrical Probability

STEP III 2007, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s3-q14

The key idea

The smallest centred circle (or square) enclosing all the holes is fixed by the *furthest* hole, so we want the distribution of a maximum, and $P(\max \le r)$ is just the nth power of the single-throw probability. On the disc, the squared distance R^2 of one hole from O turns out to be uniform on $[0,1]$, which makes everything clean. On the square, the right "distance" is $\max(|x|,|y|)$, and it behaves exactly like the radius on the disc. That coincidence also settles part (iii) almost for free.

Solution

Part (i)

Let the n holes be at distances $R_1,\dots,R_n$ from O. These are independent, since the throws are. For a single throw and $0 \le r \le 1$, the hole lies within distance r of O exactly when it lands in the disc of radius r, so

$$P(R_i \le r) = \frac{\pi r^2}{\pi \cdot 1^2} = r^2.$$

Put $U_i = R_i^2$. Then $P(U_i \le u) = P(R_i \le \sqrt{u}) = u$ for $0 \le u \le 1$, so each U_i is *uniformly distributed on* $[0,1]$, and the U_i are independent.

The smallest circle centred at O enclosing every hole has radius $M = \max_i R_i$ and area $\pi M^2 = \pi \max_i U_i$. Write $V = \max_i U_i$. By independence,

$$P(V \le v) = P(U_1 \le v)\cdots P(U_n \le v) = v^n \qquad (0 \le v \le 1),$$

so V has density nv^{n-1} and

$$E(V) = \int_0^1 v \cdot nv^{n-1}\,\mathrm{d}v = \frac{n}{n+1}.$$

Hence the expected area of the smallest circle enclosing all n holes is

$$\boxed{\frac{n\pi}{n+1}}.$$

For the circle enclosing the $n-1$ holes nearest to O, the radius is the *second-largest* distance, so the area is πW where W is the second-largest of $U_1, \ldots, U_n$.

Ask yourself How do we get the distribution of the second-largest value?

We have $W \le w$ exactly when at most one of the U_i exceeds w. Either none does (probability w^n) or exactly one does: choose which one in n ways, giving probability $n(1-w)w^{n-1}$. So

$$P(W \le w) = w^n + n(1-w)w^{n-1} = nw^{n-1} - (n-1)w^n.$$

Rather than differentiating, we can use $E(W) = \int_0^1 P(W > w)\,\mathrm{d}w$, which holds for any random variable taking values in $[0,1]$:

$$E(W) = \int_0^1 (1 - nw^{n-1} + (n-1)w^n)\,\mathrm{d}w = 1 - 1 + \frac{n-1}{n+1} = \frac{n-1}{n+1}.$$

So the expected area of the circle enclosing the $n-1$ nearest holes is

$$\boxed{\frac{(n-1)\pi}{n+1}}.$$

(This fits a pleasant pattern: n independent uniform points split $[0,1]$ into $n+1$ gaps of equal expected length $\frac{1}{n+1}$, so the kth smallest has mean $\frac{k}{n+1}$.)

Part (ii)

Take axes through Q parallel to the sides of the board, so the board is $-1 \le x, y \le 1$. We take the enclosing squares to have sides parallel to the board's. A square centred at Q with half-side s is $\{|x| \le s,\ |y| \le s\}$, so it contains the hole (x,y) exactly when $\max(|x|,|y|) \le s$.

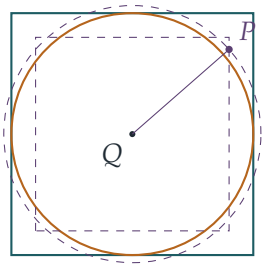
Let $S_i = \max(|x_i|, |y_i|)$ for the ith hole. For $0 \le s \le 1$,

$$P(S_i \le s) = \frac{(2s)^2}{2^2} = s^2,$$

the same as $P(R_i \le r) = r^2$ in part (i). The smallest enclosing square has half-side $\max_i S_i$ and area $(2\max_i S_i)^2 = 4\max_i S_i^2$. Exactly as in part (i), the S_i^2 are independent and uniform on $[0,1]$, so their maximum has expectation $\frac{n}{n+1}$. The expected area of the smallest enclosing square is

$$\boxed{\frac{4n}{n+1}}.$$

Part (iii)



Now the board is the square, and we want the smallest circle centred at Q enclosing every hole. A hole $P = (x, y)$ is at distance $\sqrt{x^2 + y^2}$ from Q, and

$$\sqrt{x^2 + y^2} \;\geq\; \max(|x|, |y|) = S,$$

with equality only when P lies on an axis. (In the figure, the dashed circle through P is never smaller than the inscribed circle of the dashed square through P.) So if D is the largest distance of a hole from Q, then $D \geq \max_i S_i$, and the area of the enclosing circle satisfies

$$\pi D^2 \;\geq\; \pi \max_i S_i^2.$$

From part (ii), $\max_i S_i^2$ has exactly the same distribution as $\max_i R_i^2$ on the circular board, so

$$E(\pi D^2) \;\geq\; \pi\, E(\max_i S_i^2) = \frac{n\pi}{n+1},$$

which is the circular-board answer from part (i). The inequality is strict: with positive probability no hole lies on an axis, and then $D > \max_i S_i$. So the expected area of the circle on the square board is **larger**.

Intuitively: the square board contains the circular one (the disc is its inscribed circle), and the darts spread out into the corners, so the furthest hole tends to be further away.

Checks

- For $n = 1$ the formulae give $\frac{\pi}{2}$ and 2: the mean of a uniform R^2 is $\frac{1}{2}$, so that is right. As $n \to \infty$ the areas tend to π and 4, the whole boards, as they should.
- A Monte Carlo simulation with 2×10^5 trials for $n = 2, 3, 5$ matched $\frac{n\pi}{n+1}$, $\frac{(n-1)\pi}{n+1}$ and $\frac{4n}{n+1}$ to about three significant figures; for example, $n = 3$ gave 2.3561 against $\frac{3\pi}{4} \approx 2.3562$.
- For part (iii), a quadrature of the exact distribution gave, for $n = 3$, an expected area of about 3.257 on the square board against 2.356 on the disc: larger, as claimed.

Common slips

- Working with $E(M)$ and then squaring: the expected area is $\pi E(M^2)$, not $\pi\big(E(M)\big)^2$.
- Taking the distance from the centre to be uniform on $[0,1]$. Area-proportional hitting makes R^2, not R, uniform; the outer rings are bigger.
- For the second-largest, forgetting the factor n for which hole is the one outside the circle.

Beyond STEP

The maximum and second-largest of a sample are *order statistics*. For n independent uniform variables on $[0,1]$, the kth smallest has a Beta distribution with mean $\frac{k}{n+1}$. Part (iii) is an example of *stochastic dominance* (coupling): if one random variable is always at least another, so is its expectation, and the comparison needs no detailed calculation. Grimmett and Welsh, *Probability: An Introduction*, and Blitzstein and Hwang, *Introduction to Probability*, both treat order statistics and transformations of random variables.

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